

6.2.

$$V = \text{Span} \{ (2, 1, 0, 1)^T, (1, 0, 1, 2)^T \} \subseteq \mathbb{R}^4$$

$$f((2, 1, 0, 1)^T) = (4, 1, 2, 5)^T \quad f((1, 0, 1, 2)^T) = (2, 2, -2, -2)^T$$

$$\text{báze } B = ((2, 1, 0, 1)^T, (1, 0, 1, 2)^T)$$

$$[f]_B^B = ([f(2, 1, 0, 1)]_B \mid [f(1, 0, 1, 2)]_B) =$$

$$= \left(\left[\begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} \right]_B \mid \left[\begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix} \right]_B \right) = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$$\det([f]_B^B - \lambda I_2) = (1 - \lambda)(-2 - \lambda) - 4 = \lambda^2 + \lambda - 6 = (\lambda + 3)(\lambda - 2)$$

$$\lambda_1 = -3 \quad \lambda_2 = 2$$

$$M_3: \left(\begin{array}{cc|c} 4 & 2 & 0 \\ 2 & 1 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad M_3 = \text{Span} \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\}$$

$$\text{báze } C = \left(\begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)$$

$$M_2: \left(\begin{array}{cc|c} -1 & 2 & 0 \\ 2 & -4 & 0 \end{array} \right) \sim \left(\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right) \quad M_2 = \text{Span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

$$[C_1]_B = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$[C_2]_B = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\vec{v} = \begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = 1 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + 3 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = 1 \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + 3 \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \quad [\vec{v}]_B = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix} = (-1) \begin{pmatrix} 1 \\ -2 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$[f]_C^C = \left([f\left(\begin{pmatrix} 1 \\ -2 \end{pmatrix}\right)]_C \mid [f\left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}\right)]_C \right) = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$$

$$[\vec{v}]_C = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\text{je-li } \begin{pmatrix} 1 \\ -2 \end{pmatrix} \text{ je v.l.v. } \lambda = -3 \Rightarrow f\left(\begin{pmatrix} 1 \\ -2 \end{pmatrix}\right) = -3 \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \text{ je v.l.v. } \lambda = 2 \Rightarrow f\left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}\right) = 2 \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$([f]_C^C)^n = \begin{pmatrix} (-3)^n & 0 \\ 0 & 2^n \end{pmatrix} \quad ([f(\vec{v})]_C)^n = [f^n]_C^C \cdot [\vec{v}]_C = \begin{pmatrix} (-3)^n & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -(-3)^n \\ 2^n \end{pmatrix}$$

prevedeme do báze B

$$[id]_B^C \cdot \begin{pmatrix} -(-3)^n \\ 2^n \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} -(-3)^n \\ 2^n \end{pmatrix} = \begin{pmatrix} -(-3)^n + 2^{n+1} \\ 2 \cdot (-3)^n + 2^n \end{pmatrix} = [f^n(\vec{v})]_B$$

prevedeme do kanon. báze

$$f^n(\vec{v}) = (-(-3)^n + 2^{n+1}) \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + (2 \cdot (-3)^n + 2^n) \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \cdot 2^n \\ -(-3)^n + 2^{n+1} \\ 2 \cdot (-3)^n + 2^n \\ 3 \cdot (-3)^n + 2^{n+2} \end{pmatrix}$$

$$-2 \cdot (-3)^n + 2^{n+2} + 2 \cdot (-3)^n + 2^n = 2^{n+2} + 2^n = 2^n(2^2 + 1) = 2^n \cdot 5$$